Lecture 1.2: Introduction

What is computer vision?

↳ Replicating the human eye in a machine

Examples

↳ Detect/segment images

↳ From a collection of 2D images, reconstruct 3D structure of the image

↳ Produce videos/images from different views.

↳ What is where by looking.

Computer Vision VS Biological Vision:

↳ Biological vision serves as a motivation

↳ Over 50% of pre-processing in human brain is dedicated to visual information.

Computer Vision VS Computer graphics:

↳ Given Objects, material, pose, Motion, semantic, 3D pose, shape/geometry we can concerned with generating a photorealistic 2D image of that scene.

↳ Computer vision is reverse. From pixel matter get above.

↳ It is an ill posed inverse problem

↳ Many 3D scenes yield same 2D image

↳ Additional knowledge (constraints) is required

Computer Vision VS Image processing:

↳ IP is concerned with low level manipulation of images like edge detection, color adjustment, denoising, morphing, warping etc.

↳ CV deviated from IP to get a holistic scene understanding.

Challenges:

↳ Computer sees numbers

↳ Images are 2D representation of 3D world. So, information is lost.

↳ Perspective illusion

↳ Viewpoint variations
↳ Deformation
↳ Occlusion
↳ illumination
↳ Motion
↳ Perception VS Measurement
↳ local ambiguities
↳ Variation in object themselves. Eg: Diff models of chair

Lecture 1.3: History of Computer Vision

1957: Stereo

↳ Given two images taken from aerial perspective
↳ Correlate features b/w these images
↳ To create Digital elevation maps
↳ Analog implementation.

1958 – 1962: Rosenblatt's Perceptron

↳ Invented perceptron algorithm
↳ $L(w) = -\sum_{n \in M} w^T x_n y_n$
↳ Able to classify simple patterns like male VS Female

1963: Larry Robert's Blocks World

↳ Scene understanding for robots
↳ Interpret image as projections of 3D scenes & recover the 3D Structure of an object from topological structure of lines.
↳ Taking simple images
↳ Extracting edges
↳ Coming up with a 3D fixed, uniform shapes that can be rotated
↳ Car represented as a block & spheres for wheels

1966: MIT Summer Vision project

↳ Understimated the field of CV & tried to solve it with interns

1969: Minsky and Papert publish book

↳ Several discouraging results of single-layer perceptrons
↳ Research declined

1970: MIT copy demo

↳ Culmination of block world's perception
↳ Use block worlds perception algorithm to recover the structure of a simple scene
↳ Robot plans & copy from another set of blocks
↳ Was NOT robust enough

1973: Shape from shading

↳ Recover 3D from a single 2D image
↳ Look at shading of the image

1978: Intrinsic images:

↳ Decomposing an image into it's intrinsic components such as reflectance, shading, shape and motion components

↳ Useful for downstream tasks, eg: object detection independent from lighting & shadows

1980: Photometric stereo

↳ Recover 3D from multiple 2D images, taken from the same viewpoint with different lighting conditions.

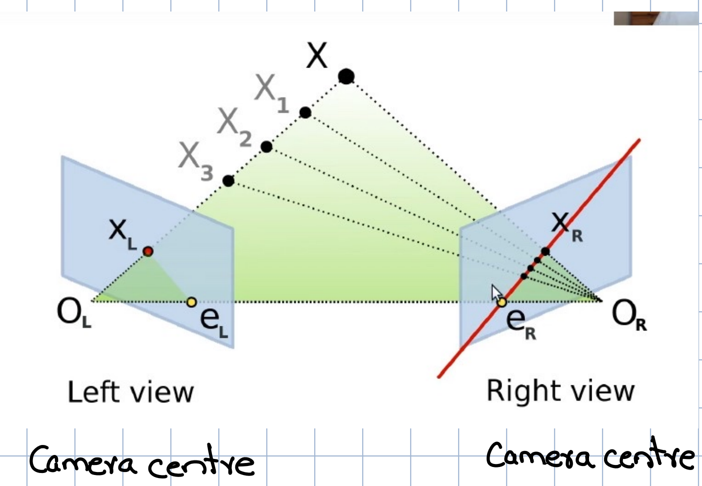
↳ Requires atleast 3 images

↳ Assumption: Object's surface were homogeneous and lambertian

1981: Essential Matrix

↳ Defines two view geometry as matrix
↳ Maps points to a epipolar lines
↳ Search along 1D line to search for X_L in order to be able to triangulate it to reconstruct 3D geometry of that point.

↳ Essential matrix defines relation b/w to images
↳ We find the epipolar line using essential matrix.

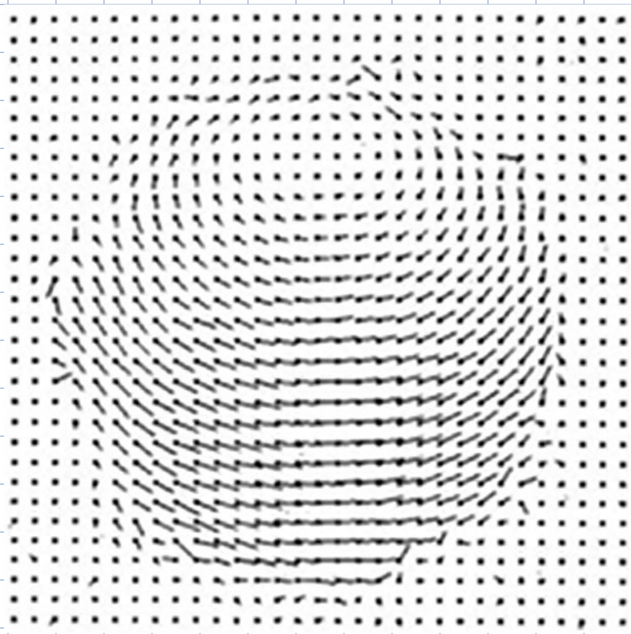


1981: Binocular Scanline Stereo

- Method for extracting depth images from 2 images from different points.
- Also construct 3D using epipolar geometry
- Points looked similar despite not being same due to ambiguity so new constraints were added to address this
- Dynamic programming was used to introduce constraints along image rows (scan lines)

1981: Dense optical flow

- It's a 2D search problem
- Doesn't assume the scene remains static
- Optical flow refers to the pattern of apparent motion of objects, surfaces & edges in a scene
- Determines how far each pixel has moved between two frames



1984: Markov Random fields

- Used to smooth images, label
- Each pixel is influenced by it's neighbors, and only neighboring pixels matter

1980s: Part based model

- 1973: Pictorial structures
- 1976: Generalized Cylinders
- 1986: Superquadrics

↳ complex scenes can decomposed into flexible, distinct parts with varying sizes & arrangement.
↳ Very few bits
↳ This also in robotics where geometric relations b/w diff parts
↳ Car represented as blocks of different sizes for body, cylinders for wheels, and other shapes for windows & head lights

1986: Backpropogation:

↳ Efficient calculation of gradients in a DNN
↳ Known in 1961 but first success in 1986.

1986: Self driving Car VaMoRs

↳ Speed: 0 – 36 km/h
↳ Different modules for preception & controls

1988: Self Driving Car ALVINN

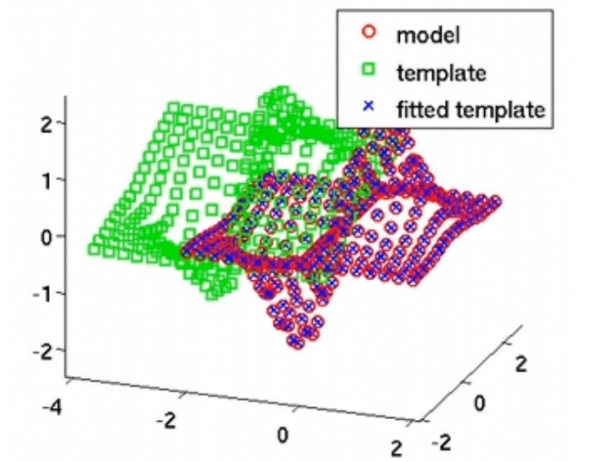
↳ Used NN
  ↳ Took input image
  ↳ Pass through NN
  ↳ Output controls
↳ In simple scenarios, able to go upto 70 mph

1992: Structure from motion

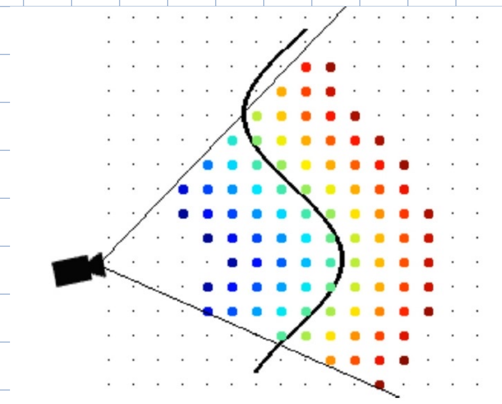
↳ Estimating 3D structures from 2D image sequences of static scenes
↳ Required only single camera
↳ Provided closed form (SVD based) solution for orthographic case (Object size remains same regardless of distance)

1992: Iterative closest point

↳ The algo starts at green template
↳ Finds the closest point b/w model & template
↳ Transforms green to minimize distance b/w matched points
↳ Repeat till convergence (blue Xs) or max repetition reached



1996: Volumetric Fusion



- ↳ 3D info from multiple viewpoints
- ↳ Consider surfaces as a whole
- ↳ Blue points infront & red behind of the surface
- ↳ Averages these observations & fuses multiple partial views of an object in a coherent 3D object.

1998: Multi-View stereo

- ↳ 3D reconstruction from multiple 2D images of an object
- ↳ It employs level set methods, which are mathematical techniques for tracking surfaces & shapes
- ↳ Unlike simple image matching, it focuses on full 3D reconst-ruction
- ↳ It accounts for which parts of the object are visible from each camera viewpoint
- ↳ Proved convergence
- ↳ flexiable topology

1998: Stereo with graph cuts

- ↳ Treats pixels as a network of graphs, where each pixel is connected to it's neighbour.
- ↳ Aims to find best overall solution for depth estimation across the entire image.
- ↳ Considers both individual pixel info (Unary terms) & how neighbouring pixels are related.

1998: Convolutional Nueral Networks

- ↳ 

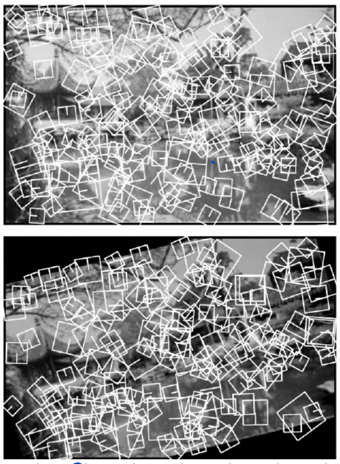
1999: Morphable Models:

- ↳ Single view 3D face reconstruction
- ↳ Linear combination of 200 Laser scans of faces
- ↳ Fitted in terms of geometry & appearence on real images

1999: SIFT

- ↳ Scale invariant Feature transformation

↳ Detection & description of salient local features in an image.
↳ Detected same features b/w two same images from different viewpoint
↳ Solved correspondence problem
↳ Enabled image stitching, reconstructions.



2006: Photo Tourism

↳ Largescale 3D reconstruction
↳ Key ingredients: SIFT feature matching, bundle adjustment

2007: PMVS

↳ Patch based multi stereo
↳ Robust reconstruction of various small & large objects

2009: Building rome in a day:

↳ 3D model of landmarks from unstructured internet photo collections and NOT images from a single camera

2011: Kinect

↳ Active light 3D sensing
↳ ML for 3D pose estimation

2009 – 2012: ImageNet & AlexNet

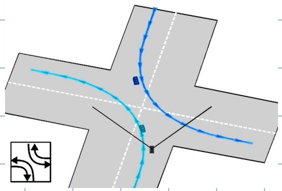
↳ CNN use in image classification

2014: DeepFace

↳ Face recognition on human level
↳ classical techniques + DNNs

2014: 3D scene understanding:

↳ holistic scene understanding
↳ Parsing RGB & Depth RGB images into holistic 3D scn understanding



2014: 3D scene recognition

↳ 3D scannig techniqes allows for creating accurate replicas

2015: Deep Reinforcement Learning

↳ Learning policy (state → action) through random exploration & reward signals

↳ Human level control on video games

2017: Style transfer

↳ one image in style of another

2015 – 2017: Semantic Segmentation

↳ Assign semantic class on every pixel